

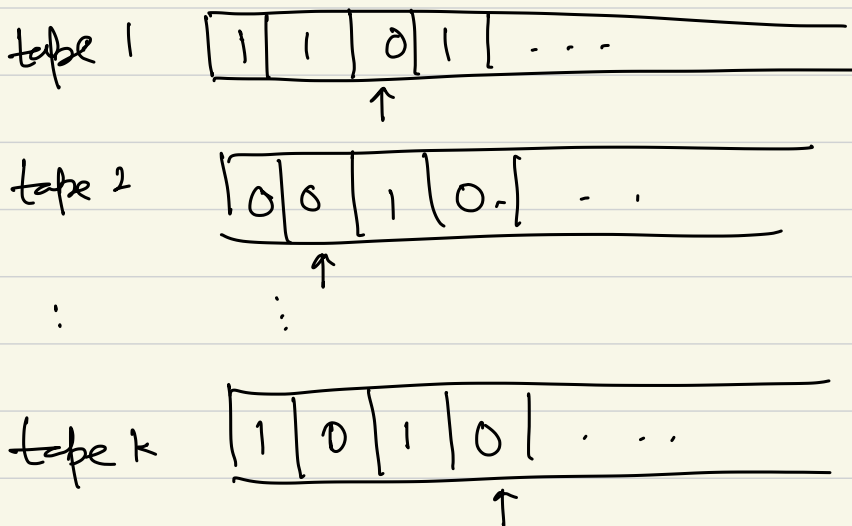
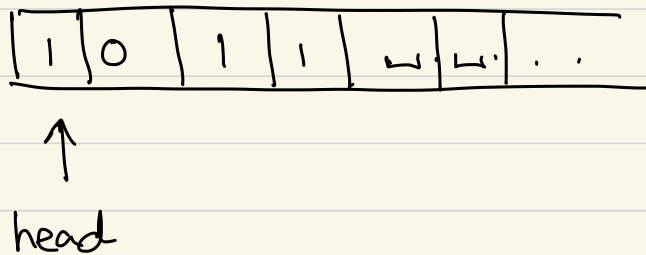
Week 7 Lecture 2. March 21, 2024



21-March-2024.

Variants of TM's

multitape TM.



$$\delta: Q \times \Gamma \rightarrow Q \times \Gamma \times \{L, R\} \quad \text{Single-tape}$$

$$\delta: Q \times \Gamma^k \rightarrow Q \times \Gamma^k \times \{L, R, S\}^k$$

$$\delta(q_i, a_1, a_2, \dots, a_k) = (q_j, \overbrace{b_1, b_2, \dots, b_k}^{\text{k-modifications of the tape}}, \underbrace{L, R, R, \dots, L}_{\text{k-movements}})$$

Theorem: Every multitape TM has an equivalent single tape TM.

Proof:

